

AI-Powered Tools for AI-Detection:

Overview and Summary

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Executive Summary

This project addresses the challenges of AI-generated content, including misinformation and privacy risks, by developing a browser extension that detects AI content and protects user privacy. The extension uses advanced AI algorithms to identify AI-generated content in real-time, offering features such as customizable detection sensitivity and a Privacy Mode to minimize data collection, ensuring users have maximum control over their experience (Kaur et al., 2024; Heidt, 2024).

The development process began with research on AI detection challenges and the creation of prototypes to explore potential solutions (Miller, 2024). A fully functional extension was developed using Chrome APIs, allowing for early user feedback. The feedback highlighted areas for improvement, including the need for clearer instructions and tutorials (CTE, 2024).

Looking ahead, future work will focus on enhancing detection accuracy to minimize false positives and negatives (Sala, 2024), integrating the extension with mobile apps and additional browsers, and linking users to educational resources to raise awareness of malicious AI use (Kerry, 2020; Restack, 2024). This project demonstrates the potential of a simple yet effective tool for navigating the risks of AI-generated content, with strong prospects for further development and adoption.

Introduction:

Advances in artificial intelligence have led to the rise of AI-generated content like deepfakes, automated videos, and synthetic text. While these technologies can be useful, they also pose risks to privacy, trust, and online security. Deepfakes, for example, can spread misinformation or impersonate people, causing harm and reducing trust (Sala, 2024). Many users don't realize when content is AI-generated, leaving a gap in awareness and tools to address these risks.

Existing tools for detecting AI-generated content are often too technical, inaccurate, or impractical for everyday use (Kaur et al., 2024). This lack of accessible solutions makes users vulnerable to misinformation and manipulation.

Our project aims to fill this gap with a browser extension that helps users identify AI-generated content and make informed decisions. Designed for ease of use and privacy, the extension includes features like customizable detection settings and minimal data collection. This tool seeks to combat misinformation, build trust, and protect user privacy in today's AI-driven digital world.

Research Question:

"How can we design an intuitive tool that helps users detect AI-generated content while protecting their privacy?"

Competitive Analysis:

Existing tools in the market address AI detection but have several limitations:

- **Gaps in Real-Time Detection:** Most tools require users to upload files to third-party apps for analysis, which is time-consuming and impractical.
- **Lack of Integration:** There are no known browser extensions or tools that seamlessly work with platforms like Instagram, YouTube, or other social media.
- **Narrow Focus:** Existing tools typically focus on detecting one type of AI-generated content, such as text, images, or audio, rather than addressing all types in a unified solution.

Our project fills these gaps by creating a real-time browser extension that integrates with social media platforms and detects multiple types of AI-generated content (text, images, and videos).

Methodology

Steps Taken:

1. **Research:** Conducted a literature review to understand challenges in AI detection and privacy.
2. **Feature Selection:** Identified key functionalities like AI detection notifications, Privacy Mode, and sensitivity adjustments.
3. **Prototyping:** Designed user flows and interfaces using Figma.
4. **Development:** Built a functional browser extension using Chrome APIs.
5. **Testing:** Collected feedback from early testers to refine the extension's usability and effectiveness.

Tools Used:

- **Figma:** Used to design visual mockups and prototypes, creating a user-friendly interface that guided the development of the browser extension.
- **Chrome API:** The development of the browser extension was powered by the Chrome API, which enabled key functionalities such as real-time notifications and privacy controls. I created the extension by building core files, including `manifest.json`, `popup.html`, `popup.css`, and `popup.js`, ensuring a seamless user experience.
- **User Feedback:** We made improvements to the app based on real-world testing and feedback from users. People liked the real-time notifications and how easy it was to adjust the sensitivity settings. However, some users mentioned that the instructions for

first-time use were unclear and suggested adding better tutorials. Using this feedback, we plan to make the app easier to use and improve the setup process in future updates.

Resulting Solution (Updated with Visuals)

Our browser extension enables users to detect AI-generated content in real-time and take control of their privacy.

Key Features:

- **Real-Time Notifications:** Alerts users when AI-generated content is detected.
- **Customizable Settings:** Sensitivity sliders, Privacy Mode, and notification preferences allow users to tailor the experience.
- **User Education:** Pop-ups link users to resources for learning about AI and its implications.



Figure 1: Mobile Prototype

This figure displays the app's home screen. Users can access the Figma prototype by clicking the app icon.

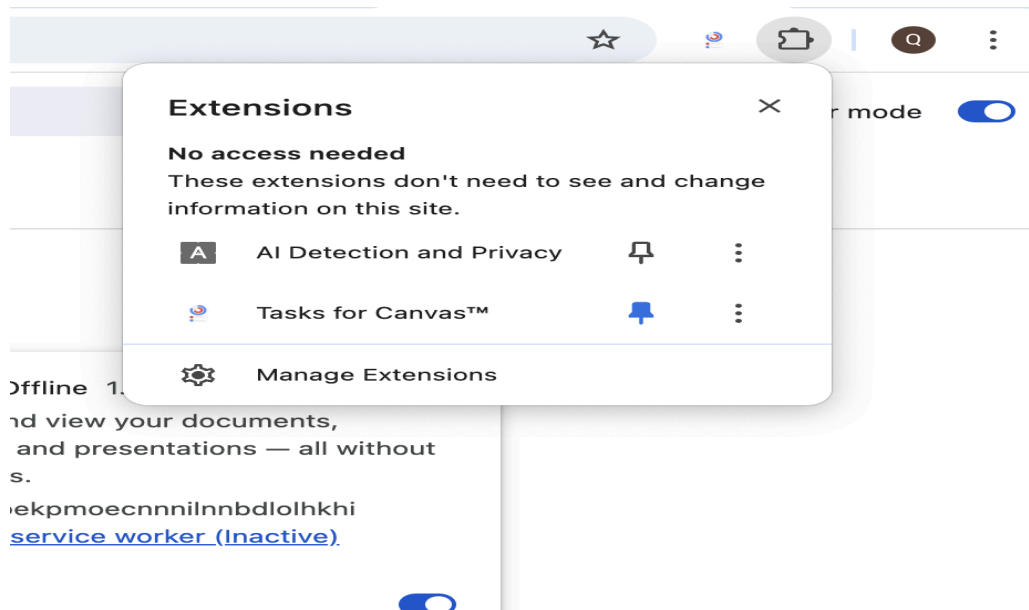


Figure 2: Browser Extension Overview

This screenshot shows the AI Detection and Privacy browser extension in action, highlighting its seamless integration into the browser's extension menu. The extension is accessible with real-time AI detection capabilities.

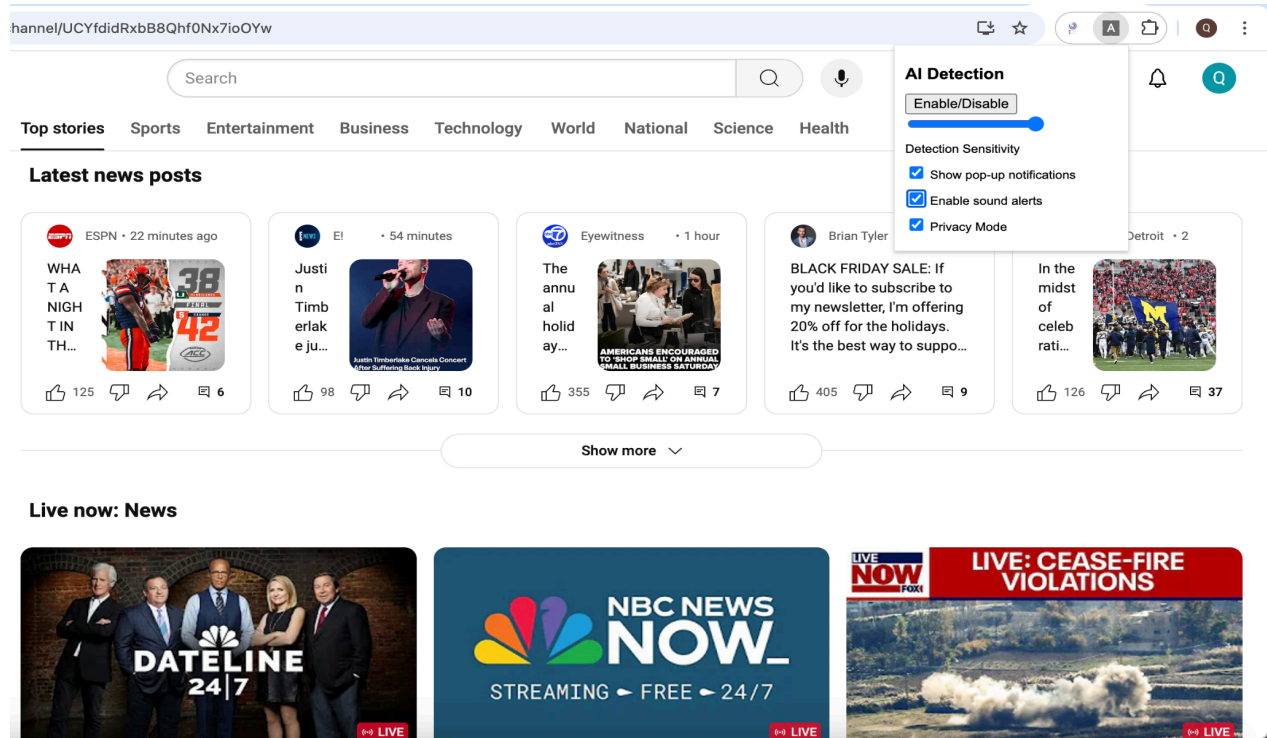


Figure 3: Browser Extension in Action

This screenshot demonstrates the AI Detection and Privacy extension actively running on a browser. The dropdown menu shows key features, including enabling/disabling detection, sensitivity adjustment, and Privacy Mode, while users browse live news content.

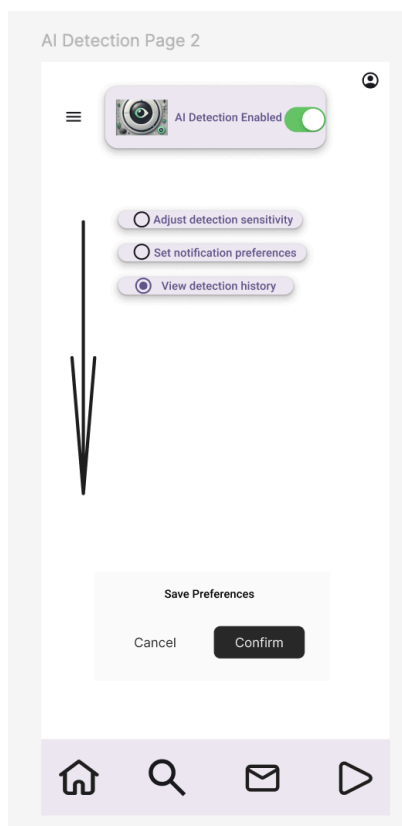


Figure 4: AI Detection Settings Page

This screenshot shows the settings interface of the AI Detection and Privacy extension. Users can adjust detection sensitivity, configure notification preferences, and view detection history. Options to save or cancel changes ensure customization and control.

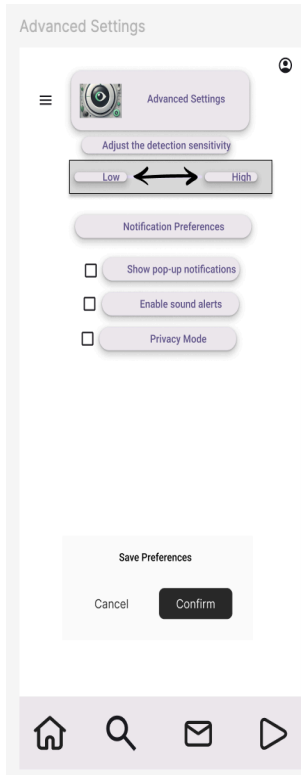


Figure 5: Advanced Settings Page

This screenshot illustrates the advanced settings interface of the AI Detection and Privacy extension. Users can adjust detection sensitivity, enable or disable specific notifications such as pop-ups or sound alerts, and activate Privacy Mode. The design ensures flexibility and control with clear options to save or cancel changes.

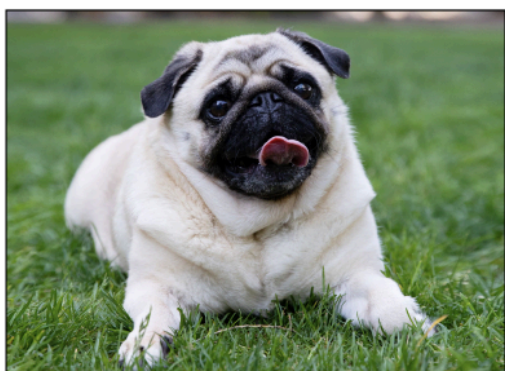
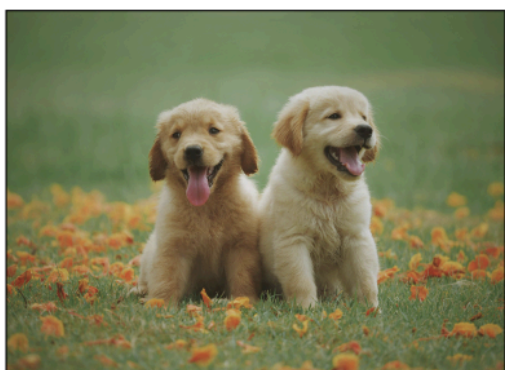
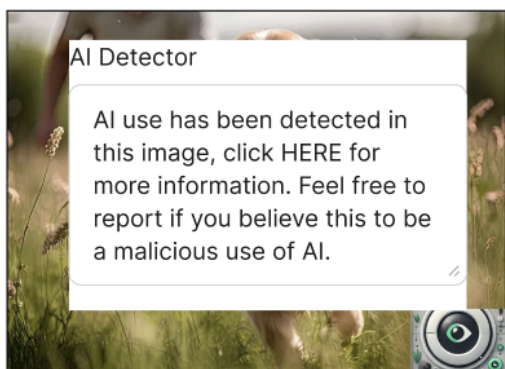


Figure 6: AI Image Detection Notification

This shows how the extension detects AI-generated images. The top image includes a notification alerting users, with a link for more details and an option to report misuse. The other images show examples of analyzed content. This feature helps users stay informed and in control of AI content.

Discussion

Our tool addresses critical challenges in AI detection and privacy by providing a user-friendly solution that promotes transparency and trust.

Impact

- **Empowering Users:** By providing real-time notifications about AI-generated content, the tool empowers users to make informed decisions about the content they consume and share. This reduces the risk of unintentionally spreading misinformation.
- **Combating Misinformation:** On platforms like **Instagram** and **YouTube**, where AI-generated content such as deepfakes or synthetic text posts can go viral, our tool provides transparency by flagging potentially misleading content. For example, a user scrolling on YouTube would receive a notification if an AI-generated video is detected, enabling them to evaluate the authenticity before engaging.
- **Enhancing Privacy:** Features like "Privacy Mode" protect users by limiting data collection while maintaining AI detection functionality. This fosters user trust in digital spaces where privacy concerns are growing.

Limitations

- **Detection Challenges:** As AI systems evolve, they may produce increasingly realistic content that is harder to detect. Our current algorithms might struggle with identifying subtle or highly advanced AI-generated materials, such as seamlessly integrated deepfakes.

- **Adoption Barriers:** Users unfamiliar with browser extensions or AI detection tools may find the onboarding process intimidating without adequate guidance. Addressing this through tutorials or pop-ups will be essential for widespread adoption.
- **Performance Constraints:** On older devices or slower browsers, the extension may impact performance, potentially leading users to disable or uninstall it. Optimizing performance across devices will be a key focus for future iterations.

Future Work:

- **Algorithm Improvements:** Enhance detection algorithms to better handle complex and emerging AI-generated content, such as voice synthesis or multi-modal media.
- **Mobile Integration:** Expand compatibility to mobile platforms, allowing direct integration with popular apps like Instagram and YouTube for seamless user experience. It would lead to more use of the app and making it run in the background would increase the efficiency of detection.
- **User Education:** Develop educational resources, such as interactive tutorials, to raise awareness about AI and privacy risks and improve user onboarding. We hope it will lead to less harm done by malicious actors online whilst also providing people with a sense of safety and security over what they are interacting with online.

Conclusion:

Our project demonstrates the feasibility of creating a simple yet effective tool for AI detection and privacy protection. By providing users with real-time notifications, customizable settings, and educational resources, our solution directly addresses critical challenges posed by AI-generated content, such as misinformation and privacy risks (Sala, 2024; Kaur et al., 2024).

The browser extension empowers users to make informed decisions about the content they consume, fostering trust and transparency in digital spaces. While the tool currently focuses on AI detection, future iterations will prioritize enhancing detection algorithms to minimize false positives and negatives, expanding compatibility with mobile apps and additional browsers, and integrating features that support user education and onboarding. These improvements will help overcome current limitations, such as detection challenges and adoption barriers, ensuring broader accessibility and effectiveness.

Ultimately, this project lays a strong foundation for navigating the complexities of AI-driven content in a rapidly evolving digital landscape. By equipping users with an intuitive and privacy-conscious tool, we aim to contribute to a safer, more transparent online environment where individuals can remain informed and in control of their experiences. This work serves as a stepping stone for future innovations in AI ethics, detection, and privacy, encouraging ongoing research and collaboration to address the emerging challenges of an AI-driven world.

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Appendices:

Raw Data From Usability Testing

Participants:

- 4 users with varying levels of tech expertise:
 - 2 beginner users
 - 2 advanced users

Testing Scenarios:

- Detecting AI-generated content on a news website.
- Adjust detection sensitivity and enable Privacy Mode.
- Interpret notifications and take further action (e.g., clicking for more information).

Tools Used: Browser extension prototype on Chrome.

Key Metrics Collected

Metric	Results
Time to Detect AI Content	Average: 4 seconds

Feature Understanding	100% successfully adjusted sensitivity settings.
Notification Comprehension	100% correctly interpreted detection pop-ups.
Privacy Mode Activation	100% successfully enabled Privacy Mode.

Participant Feedback Highlights

1. Positive Feedback:

- "The interface is clean and easy to navigate."
- "I like that I can adjust detection sensitivity based on my needs."
- "The Privacy Mode is a great feature—it makes me feel more secure."

2. Negative Feedback/Challenges:

- "I wasn't sure what some of the pop-up notifications meant at first."
- "It took me a few tries to find the sensitivity slider."
- "The extension slowed down my browser slightly on older devices."

3. Suggestions for Improvement:

- Provide a short tutorial or onboarding walkthrough for new users.
- Optimize the wording of notifications making them clearer and more actionable.